

CHEATSHEET · QUICK REFERENCE

Lab 13 — Data Migration and Transfer

Move data with rsync, scp and sftp, choose the right cross-OS tool including Robocopy, fast-copy large files and control exfiltration risk.

rsync - the workhorse

```
rsync -avh /srv/src/ /srv/dst/
Archive copy (perms/times/symlinks)

rsync -avh --dry-run --delete /srv/src/ /srv/dst/
Preview an exact-mirror sync

rsync -avh --partial --progress /srv/src/ /srv/dst/
Resume interrupted transfers

rsync -avzh -e ssh /srv/src/ user@remote:/srv/dst/
Over SSH, compressed

rsync -avh --bwlimit=1024 /srv/src/ user@remote:/srv/dst/
Bandwidth-cap to limit exfil
```

scp / sftp / fast copy

```
scp -r /srv/src/ root@127.0.0.1:/srv/dst-scp/
Simple recursive copy (no resume)

sftp root@127.0.0.1
Interactive transfer over SSH

pv /srv/src/file > /srv/dst/file
Show throughput while copying

dd if=src of=dst bs=1M oflag=direct status=progress
Direct I/O for huge migrations
```

Cross-OS transfer choices

Direction	Tool
Linux -> Linux	rsync over SSH
Windows -> Windows	Robocopy src dst /MIR /Z /R:3
Windows -> Linux	SMB mount + rsync, or WinSCP
Linux -> Windows	Samba share + Robocopy